Discrete Black-Hole Spectroscopy Does Not Force Binary Rank Without an Alphabet Law

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Abstract

Bekenstein–Mukhanov spectroscopy is a genuine possible positive route to a finite alphabet: under an integer degeneracy law $g_n = k^n$, a resolved fundamental area spacing $\Delta A = 4\log(k)\,\ell_P^2$ in fixed units would determine k, and $\Delta A = 4\log(2)\,\ell_P^2$ would determine $k = 2$. The point of this note is the converse boundary. Entropy, qualitative discreteness, recovery data, or an area spacing without the physical law identifying that spacing with an integer degeneracy base do not force the binary predicate $q = 2$. Binary rank descends from spectroscopy exactly when the supplied spectral law and measured data are injective on the binary-versus-nonbinary comparison fiber.

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1 Primary sources

The boundary below was checked against Bekenstein's entropy-area proposal [1], Bekenstein–Mukhanov quantum black-hole spectroscopy [2], Bekenstein's black-hole-as-atom review [3], and Maggiore's quasinormal-mode interpretation of area quantization [4]. These works are used as positive spectroscopy or area-quantization inputs, not as claims that binary rank is absent from nature.

2 Alphabet identifiability

Let $q \geq 2$ be the proposed finite alphabet size of a horizon cell or finite central primitive. Let $D(q)$ denote the actual spectroscopy data reported under a specified physical law: resolved fundamental frequency or area spacing, transition intensities, level degeneracies, Planck/area units, and the rule connecting those data to q.

Theorem 1 (spectroscopy alphabet criterion)**.** *The binary predicate $q = 2$ is reconstructible from the supplied spectroscopy data D if and only if, inside the declared comparison class,*

$$D(q) = D(2) \quad \Longrightarrow \quad q = 2. \tag{1}$$

Equivalently, the spectroscopy map must be injective on the binary-versus-nonbinary fiber.

Proof. If $q = 2$ is reconstructible from D, then the predicate is a function of the data, so any model with the same data as $q = 2$ must also satisfy $q = 2$. Conversely, if (1) holds, define the binary predicate on the image of D by declaring the fiber $D(2)$ binary and every other image nonbinary. Condition (1) is exactly the well-definedness condition for this assignment on the relevant comparison class. □

3 Bekenstein–Mukhanov as a positive exit

In the Bekenstein–Mukhanov idealization, the area levels are equally spaced and the degeneracy of the n-th level has the form

$$g_n = k^n, \qquad k \in \{2, 3, \ldots\}.$$

Matching the entropy $S_n = \log g_n = n \log k$ to the Bekenstein–Hawking normalization $S = A/(4\ell_P^2)$ gives

$$A_n = 4n \log(k)\, \ell_P^2, \qquad \Delta A = 4 \log(k)\, \ell_P^2. \tag{2}$$

Proposition 1 (what the line measurement supplies)**.** *Assume the Bekenstein–Mukhanov integer-degeneracy law $g_n = k^n$, fixed Planck/area units, and a resolved fundamental spacing ΔA. Then the spacing determines k exactly when*

$$k = \exp\left(\frac{\Delta A}{4\ell_P^2}\right)$$

is a positive integer satisfying the declared law. In particular, $\Delta A = 4\log(2)\, \ell_P^2$ gives $k = 2$.

Proof. Equation (2) gives $\Delta A = 4 \log(k)\ell_P^2$. Solving for k gives the displayed expression. The conclusion that k is an alphabet size uses the integer-degeneracy law; the measured spacing alone supplies only the numerical spacing parameter. □

4 What does not force binary rank

Proposition 2 (spacing without an alphabet law is nonidentifying)**.** *The following data do not by themselves imply $q = 2$:*

1. *entropy proportional to area;*
2. *qualitative discreteness of the area or mass spectrum;*

3. *a uniformly spaced area spectrum $A_n = \gamma n \ell_P^2$ with unspecified γ;*

4. *a spacing parameter γ without a law identifying $\gamma = 4\log(q)$ for an integer finite alphabet q;*

5. *recovery, sector, modular, crossed-product, or quotient-visible records that do not contain a q-separating spectral measurement.*

Proof. Entropy gives a number of states, $\exp(S)$, not a distinguished base for writing that number as powers of 2, 3, or another alphabet. A uniformly spaced spectrum with free parameter γ is compatible with $\gamma = 4\log(2)$, $\gamma = 4\log(3)$, $\gamma = 8\pi$, or other proposed area quanta, depending on the physical quantization rule. Therefore the data listed above fail condition (1): the binary and nonbinary comparison classes are not separated until the finite-alphabet law or direct measurement channel is part of the supplied data. $\square$

5 Referee question

The exact external question is:

> Does the black-hole spectroscopy or area-quantization hypothesis in the target setting supply a resolved finite-alphabet law whose data are injective on $q = 2$, or does it supply only entropy, qualitative discreteness, or area spacing without binary-alphabet accountability?

If the first holds, spectroscopy is a positive exit from the Team 1 finite central obstruction. If the second holds, Team 1 should not claim a spectroscopic derivation of binary rank; the correct conclusion is only that a direct finite-alphabet measurement would be enough.

References

[1] J. D. Bekenstein, *Black Holes and Entropy*, Phys. Rev. D 7 (1973), 2333, doi:10.1103/PhysRevD.7.2333.

[2] J. D. Bekenstein and V. F. Mukhanov, *Spectroscopy of the quantum black hole*, Phys. Lett. B 360 (1995), 7–12, arXiv:gr-qc/9505012.

[3] J. D. Bekenstein, *Quantum Black Holes as Atoms*, in Proceedings of the Eighth Marcel Grossmann Meeting, arXiv:gr-qc/9710076.

[4] M. Maggiore, *The physical interpretation of the spectrum of black hole quasinormal modes*, Phys. Rev. Lett. 100 (2008), 141301, arXiv:0711.3145.